Sri Lanka Institute of Information Technology

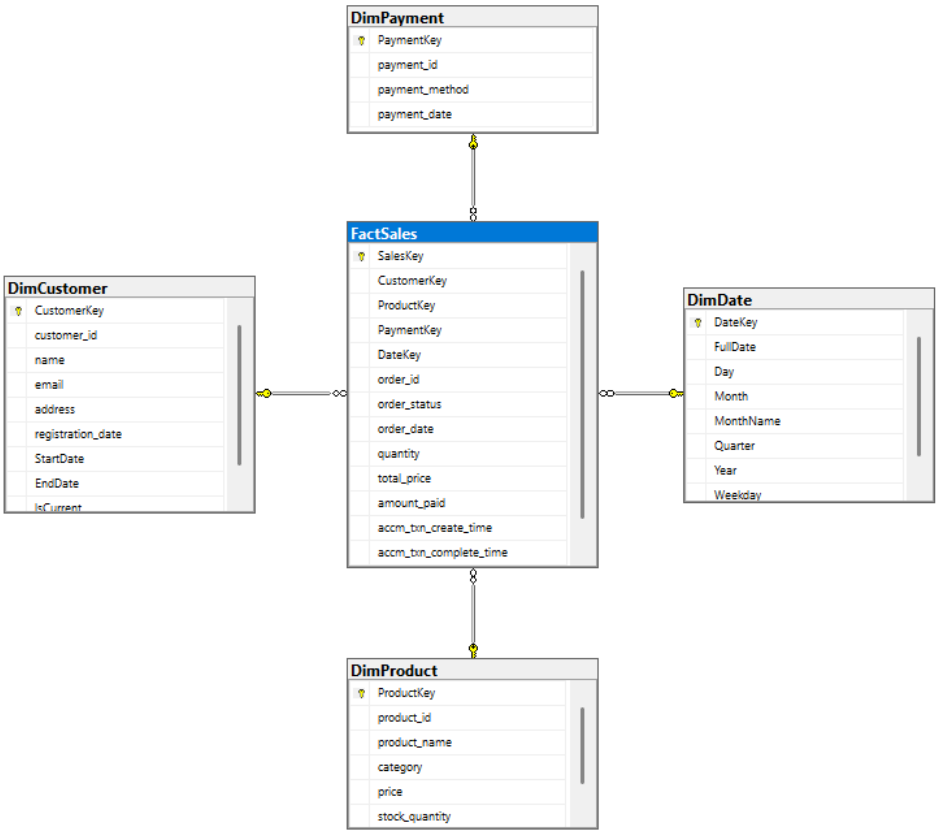
Data Warehousing and Business Intelligence IT3021

Assignment 2

DE SILVA D T N U IT23814592

Contents

1.Data source for the assignment 2



1. DimCustomer

Contains customer demographic and profile information such as name, email, gender, city, and signup date.

This dimension implements **Slowly Changing Dimension Type 2 (SCD2)** to track historical changes (e.g., city or email updates).

- **Primary Key:** CustomerKey (surrogate key)
- **SCD2 Columns:** StartDate, EndDate, IsCurrent

This ensures that historical customer data is preserved for accurate reporting.

2. DimProduct

Stores product-related information including product name, category, brand, price, and rating.

- **Primary Key:** ProductKey

This allows sales data to remain consistent with the product attributes at the time of transaction.

3. DimPayment

Captures payment-related details such as payment method and payment date.

- **Primary Key:** PaymentKey

This dimension supports analysis of payment trends and methods.

4. DimDate

A standard time dimension used across the warehouse.

It contains attributes such as:

- Day
- Month
- MonthName
- Quarter
- Year
- Weekday
- **Primary Key:** DateKey (formatted as YYYYMMDD)

This dimension supports hierarchical analysis:
Year → Quarter → Month → Day

Fact Tables

1. FactSales

This is the **central fact table** for sales analysis.

It stores key metrics such as:

- Quantity
- OrderID
- Order Date
- Order Status
- Amount_paid
- Total_price

- **Primary Key** SalesKey

It references all dimension tables using surrogate keys.

Special Feature:
Implements an **accumulating fact table pattern** with:

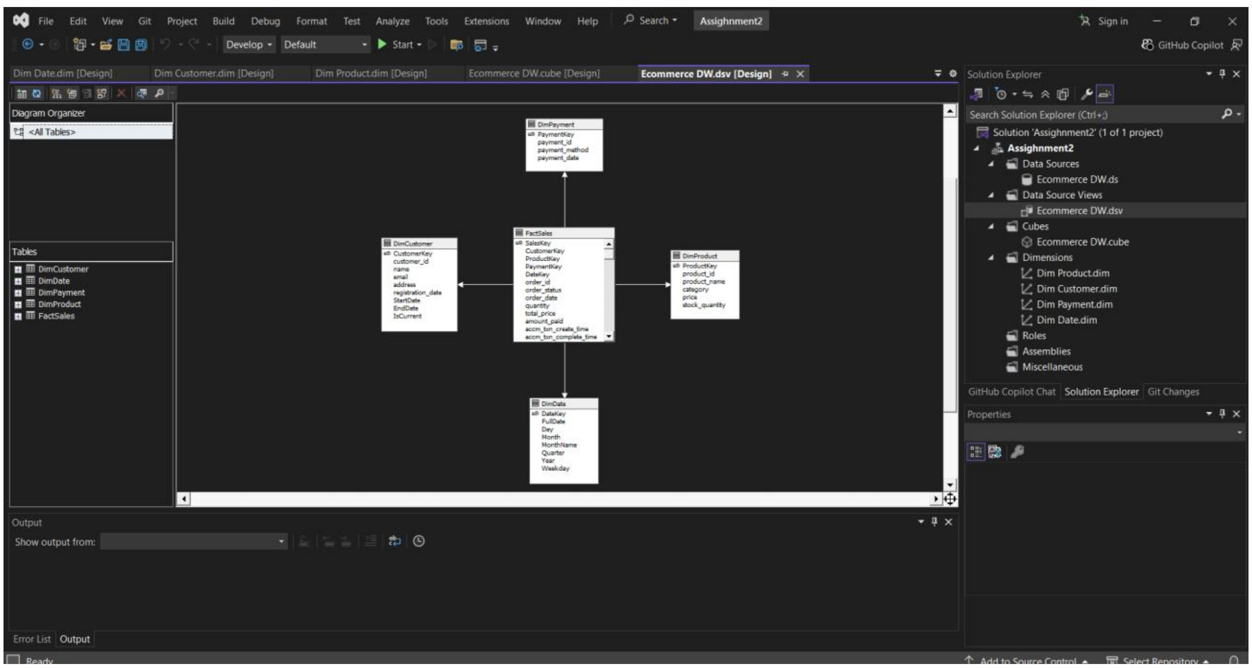
- accm_txn_create_time
- accm_txn_complete_time
- txn_process_time_hours

This tracks the full lifecycle of an order from creation to completion.

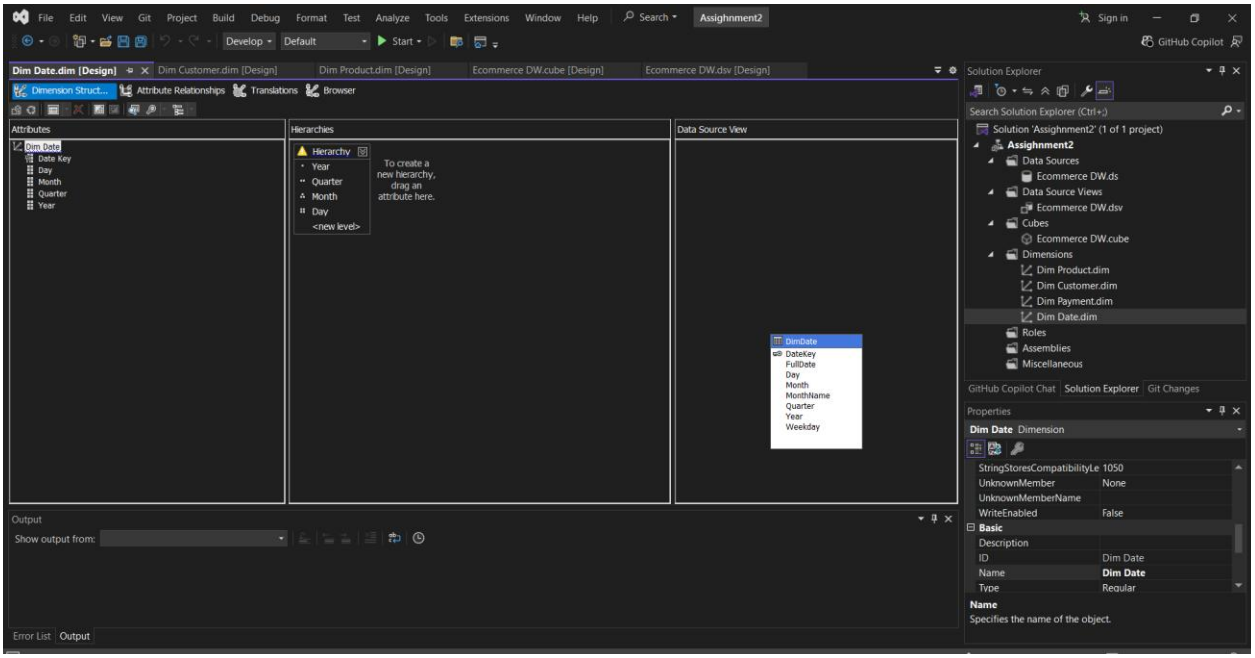
2.SSAS Cube Implementation

- Opened Visual Studio and created a new "Analysis Services Multidimensional and Data Mining Project" to start building the SSAS cube.

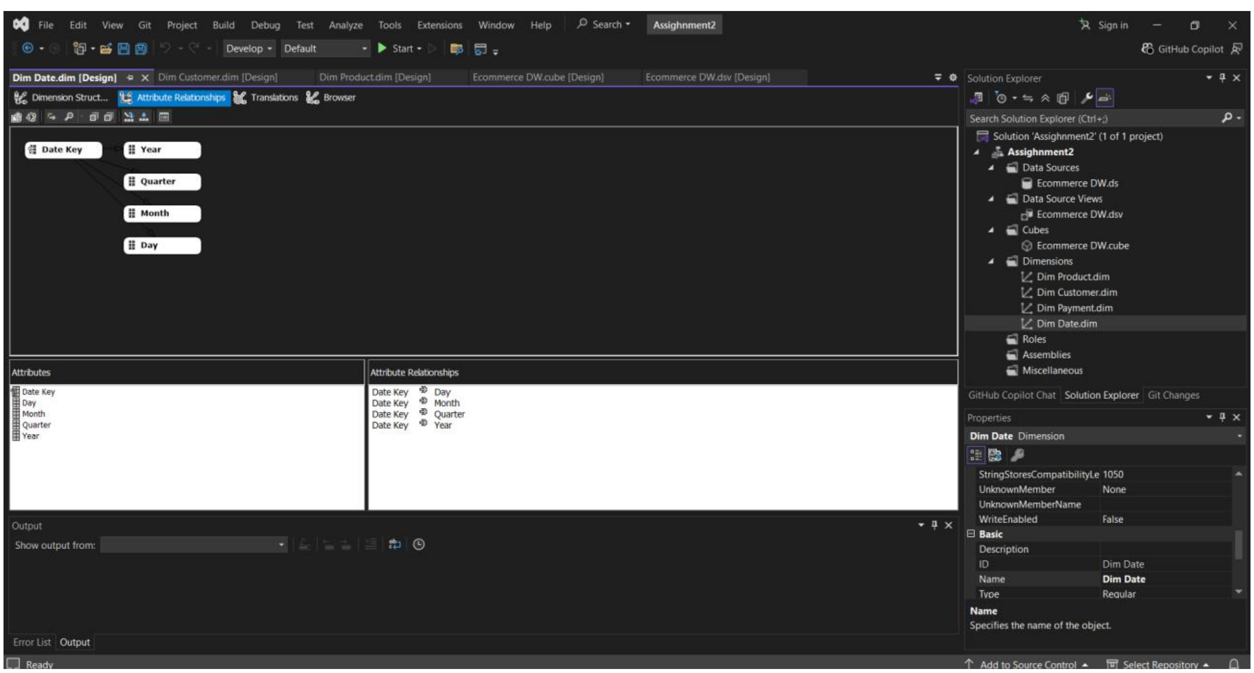
- A new Data Source View was created using the data warehouse connection. All required dimension tables and the fact table were added as included objects.



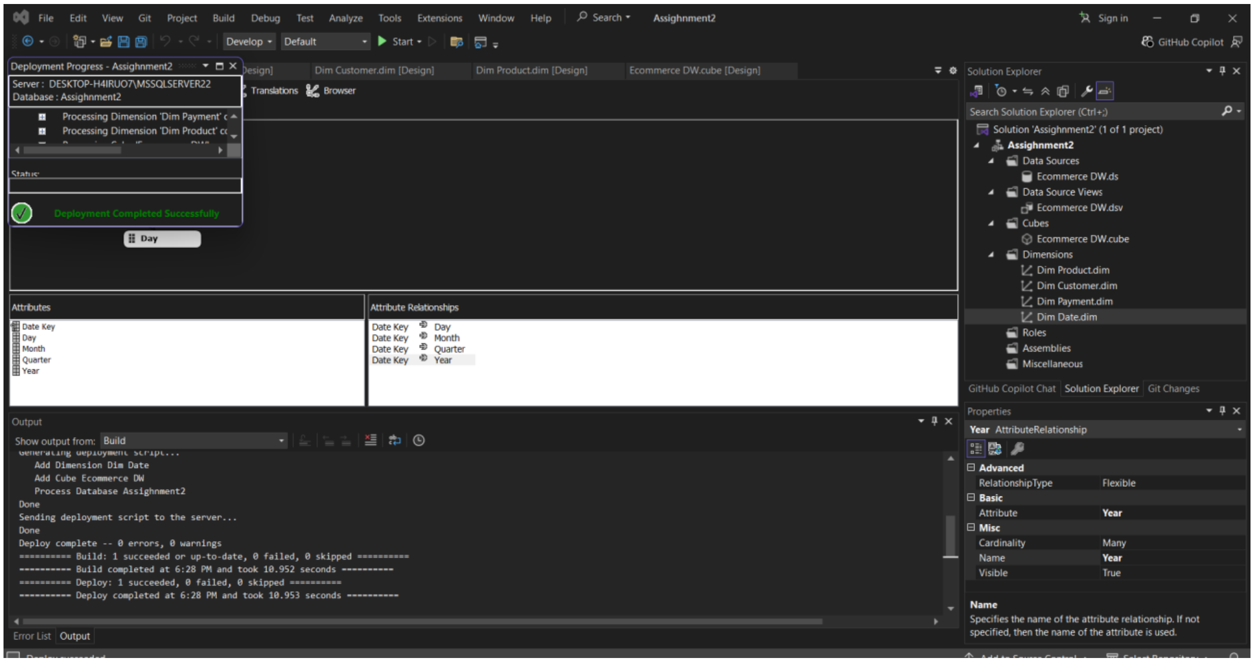
- A hierarchy was created in the Date dimension by arranging the Year, Quarter, Month, and Day attributes in logical order to support drill-down analysis.



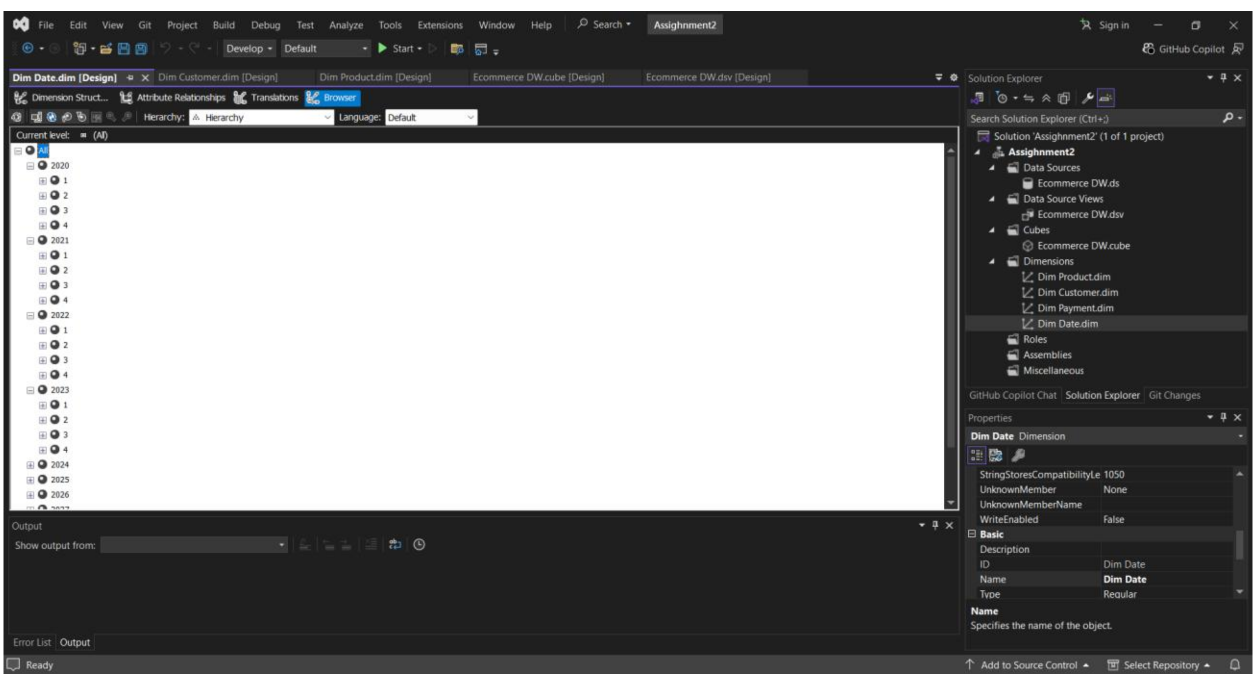
- The Date hierarchy was manually created by mapping the Year, Quarter, Month, and Day attributes in the correct order within the Date dimension structure.



- The cube was deployed to the SQL Server Analysis Services instance to create the necessary metadata and structure on the server.



- The Cube Browser was used to verify that the Date hierarchy was functioning correctly by drilling down from Year to Day, confirming proper hierarchy behavior.

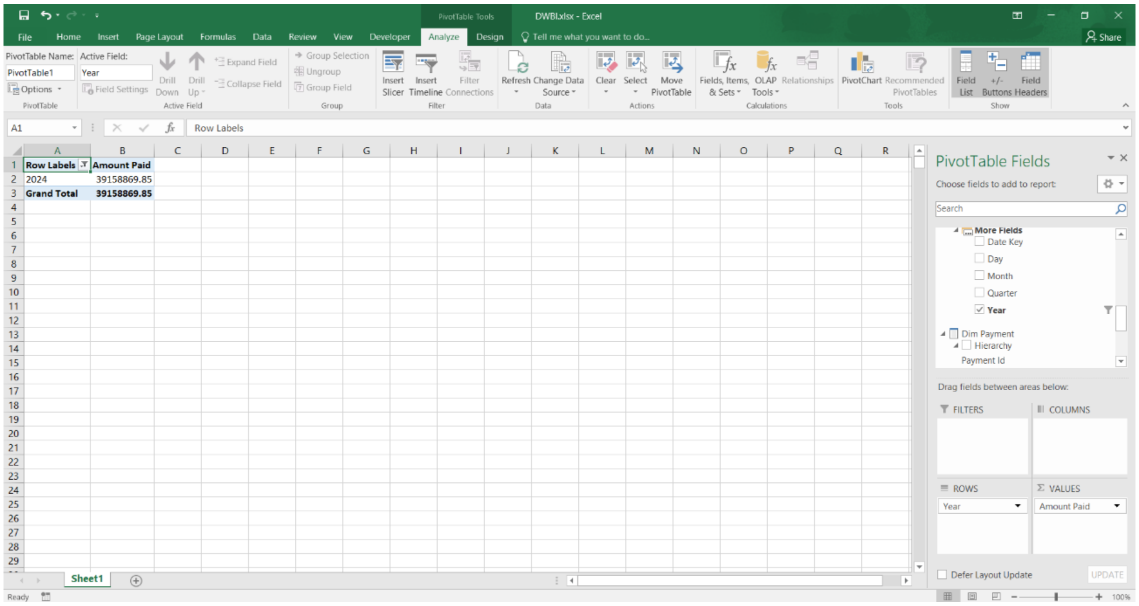


3.Demonstration of OLAP operations

Connected Excel to SSAS Cube Used Excel's Get Data feature to establish a connection to the deployed Assighment2 on the SSAS server DESKTOP-H4IRUO7\MSSQLSERVER22. Selected the Assighment2 database and inserted a PivotTable report linked to the cube.

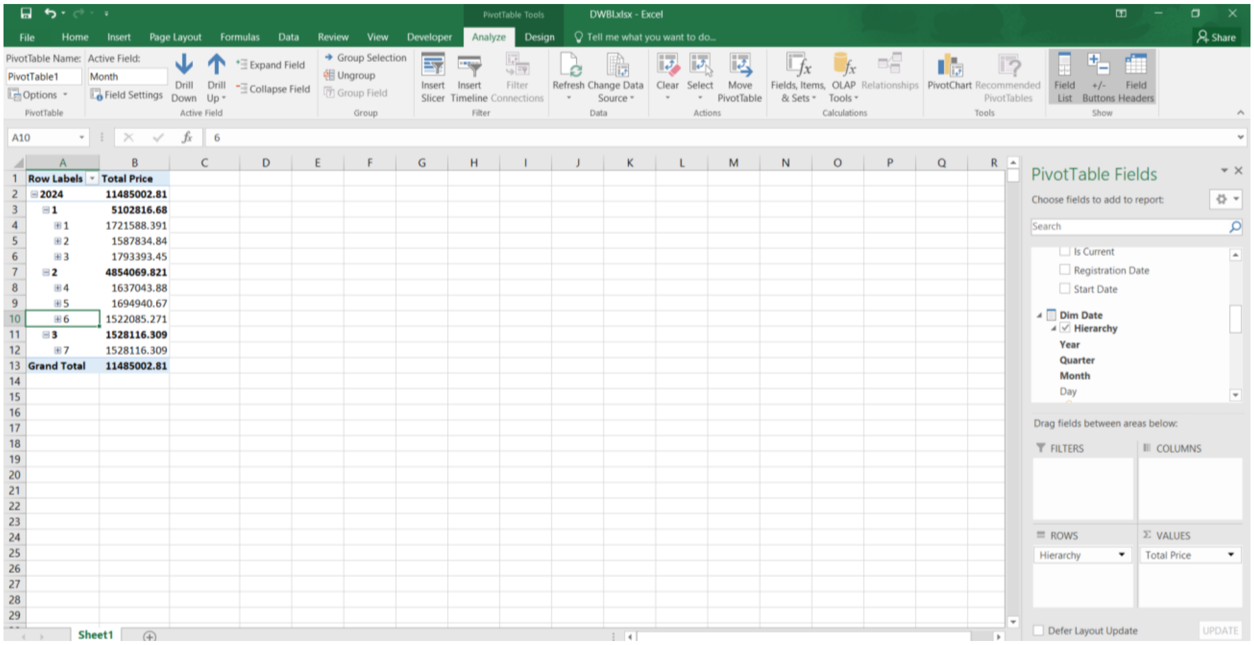
Roll- up

Roll-up Aggregated sales data from lower levels to higher levels using the Date Hierarchy. Daily transaction records were rolled up to Monthly totals, and Monthly totals were further aggregated to Yearly summaries. For example, individual order amounts across all days were summarized to show Total Amount paid per Year, demonstrating how detailed transactional data can be consolidated into high-level business summaries.



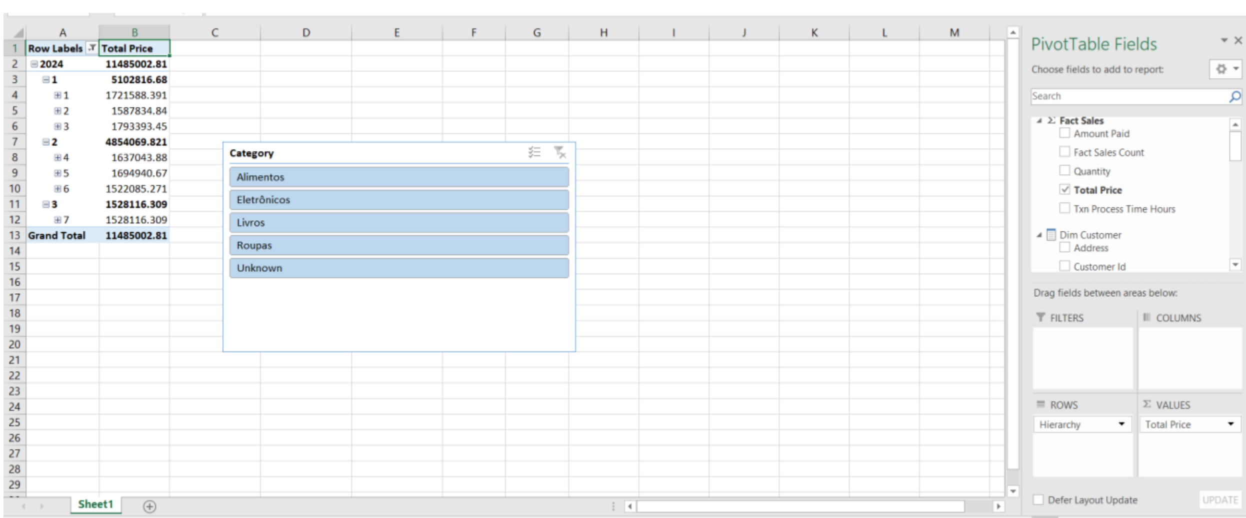
Drill-Down

Drill-down Navigated from high-level yearly summaries down to granular monthly details using the Date Hierarchy (Year → Quarter → Month → Day). By clicking the expanding (+) button next to a Year value in the PivotTable, the data was broken down into Quarters and further expanded to show individual Month-level Total Price figures. This allowed detailed examination of seasonal sales patterns within each year.



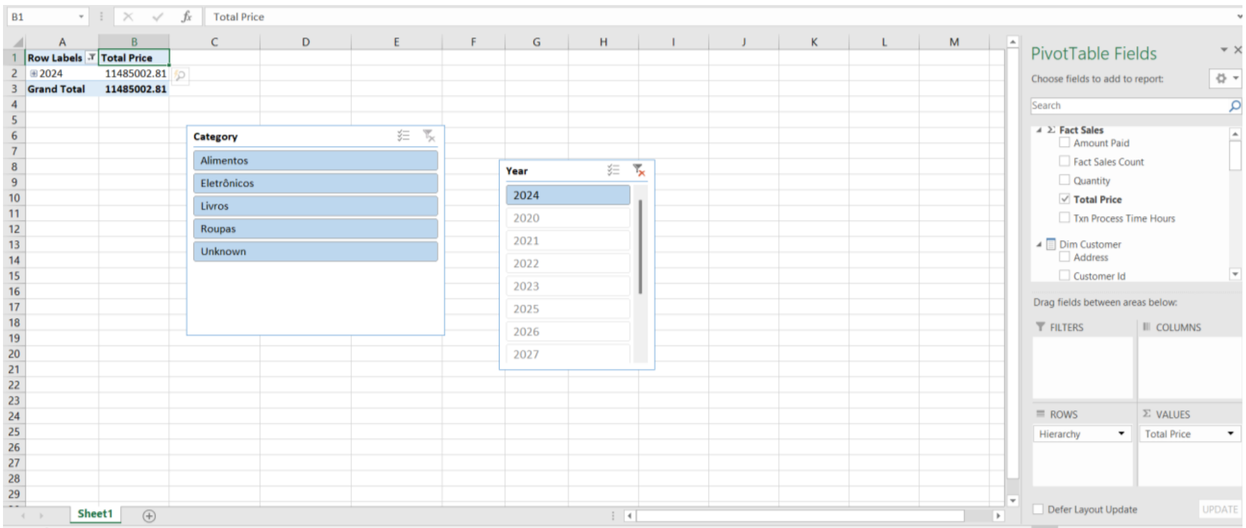
Slice

Slice Filtered the PivotTable on a single dimension value using an Excel Slicer. A slicer was inserted on the Category field from DimProduct, and a single product category (e.g. "Electronics") was selected. The PivotTable automatically filtered to display Total Revenue and order counts for Electronics products only across all time periods, isolating a single dimension for focused analysis.



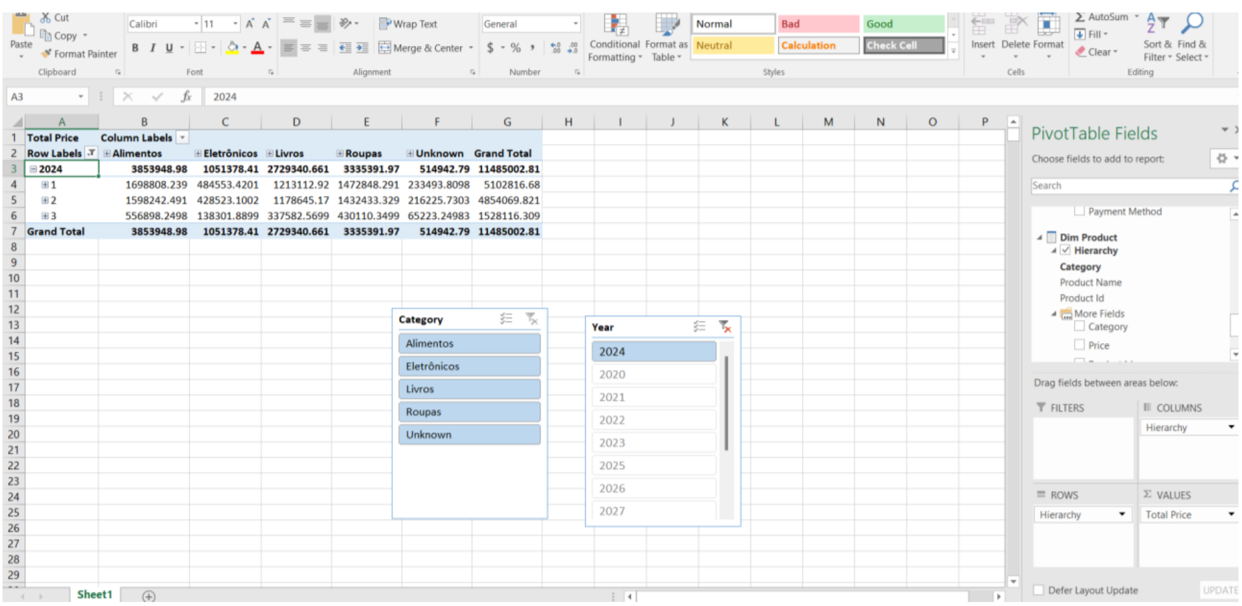
Dice

Dice Applied simultaneous filters across two dimensions to analyze a specific data subcube. Two slicers were used together — one filtering by Category (e.g. "Electronics") and another filtering by Year (e.g. "2024"). The PivotTable displayed Total Revenue only for Electronics products sold in 2024, demonstrating how multiple dimension filters can be combined to extract a precise subset of the data.



Pivot

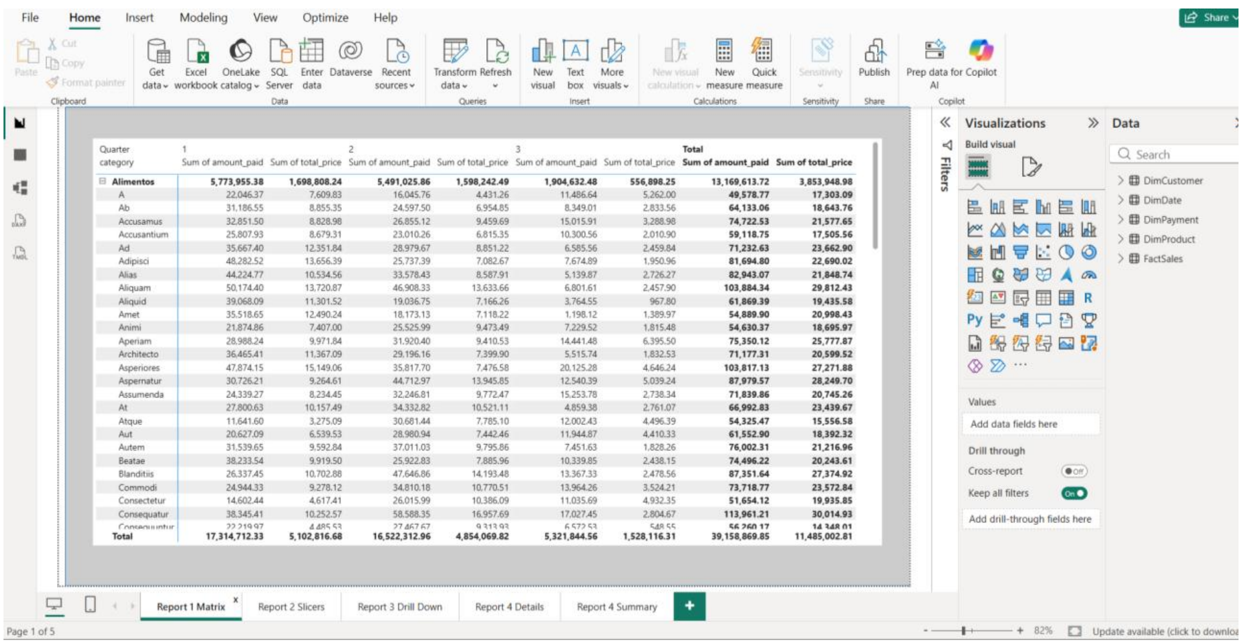
Pivot Rotated the data axes of the PivotTable to reorganize the analytical perspective. The Category dimension was moved from the Rows area to the Columns area while Year remained in Rows, resulting in a cross-tabular view where each row represents a Year and each column represents a Product Category. This rearrangement provided a different analytical viewpoint of the same underlying data without changing the data itself.



4.PowerBI Reports

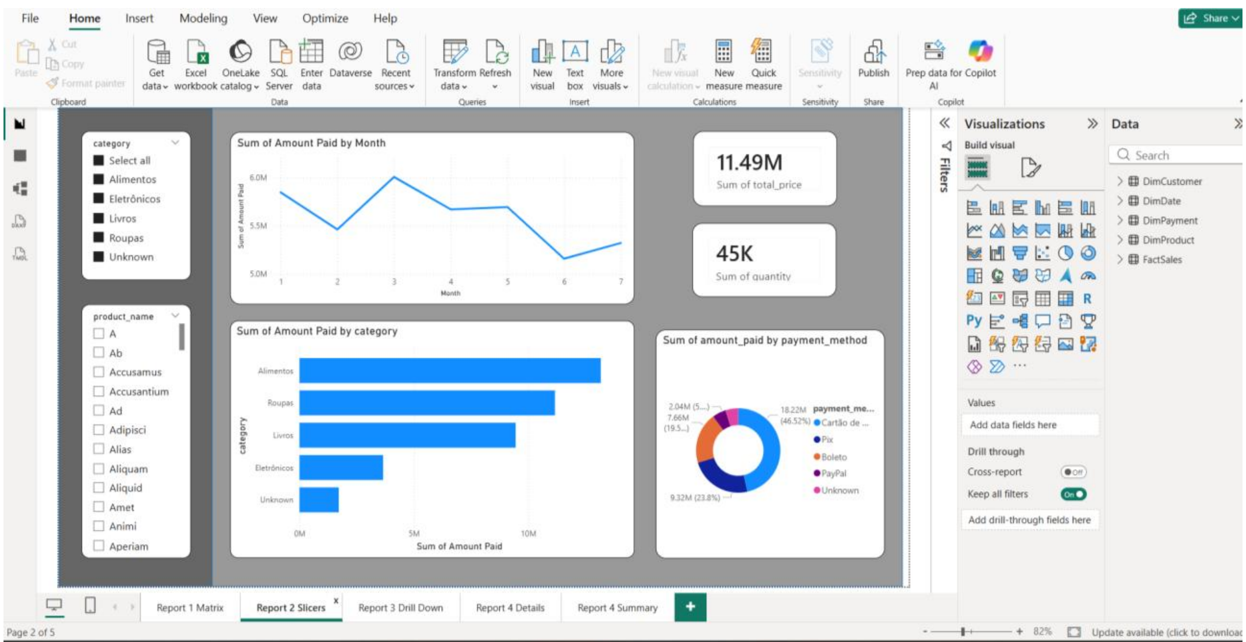
Report 1

Report 1 presents a **Matrix visual** that displays sales data with hierarchical row and column groupings. The rows are grouped by **Product Category** and **Product Name**, while the columns are grouped by **Year** and **Quarter**. The measures displayed are **Total Amount** and **Total price of product** . Row subtotals and column subtotals are enabled to provide summary values at each grouping level. This report allows users to analyze sales performance across different product categories and time periods in a structured tabular format, making it easy to compare sales figures across multiple dimensions simultaneously.



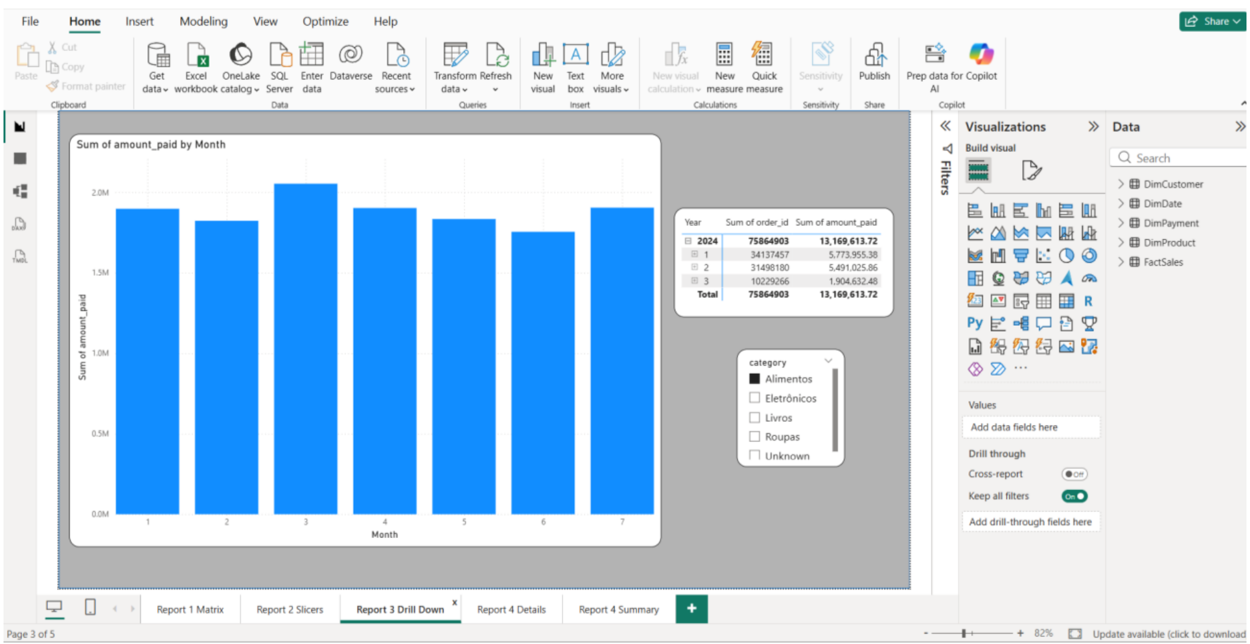
Report 2

Report 2 demonstrates **cascading filters** using two interactive slicers. The first slicer filters by **Product Category** and the second slicer dynamically updates to show only the **Product Names** belonging to the selected category. This cascading filtering behavior ensures that users can narrow down their selection in a logical top-down manner. The report also includes a **Bar Chart** showing Total Amount by Category and a **Line Chart** showing Total Amount by Month. When a category is selected in the first slicer, all visuals on the page including both charts and the product slicer are automatically filtered to reflect only the relevant data for that selection.



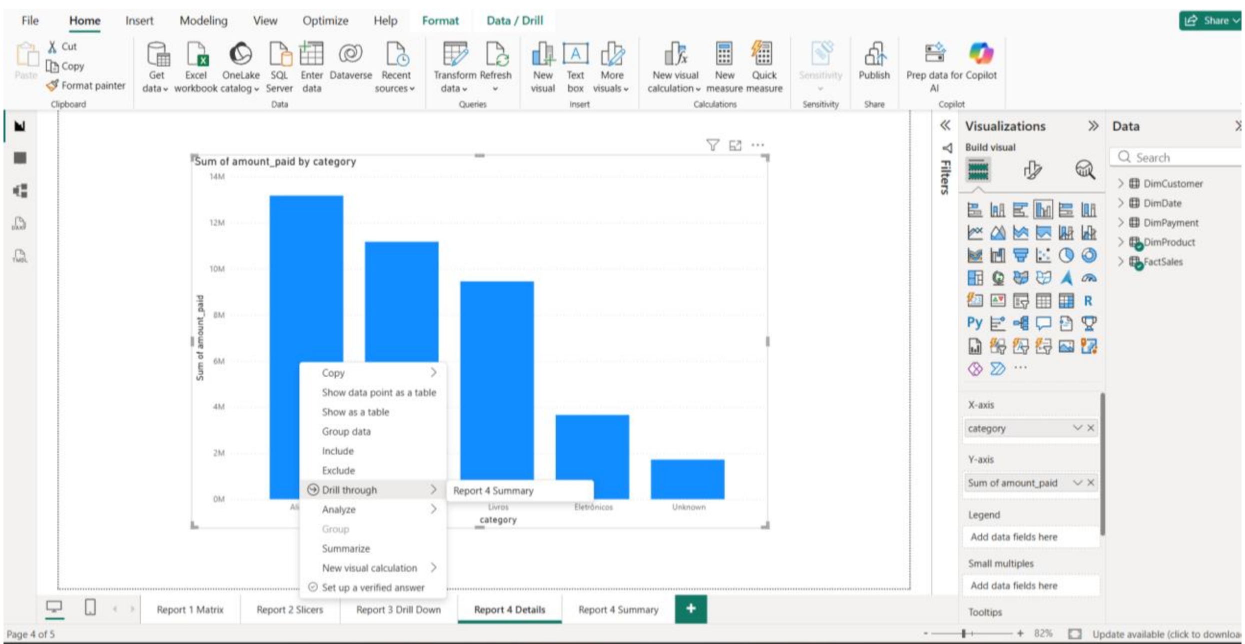
Report 3

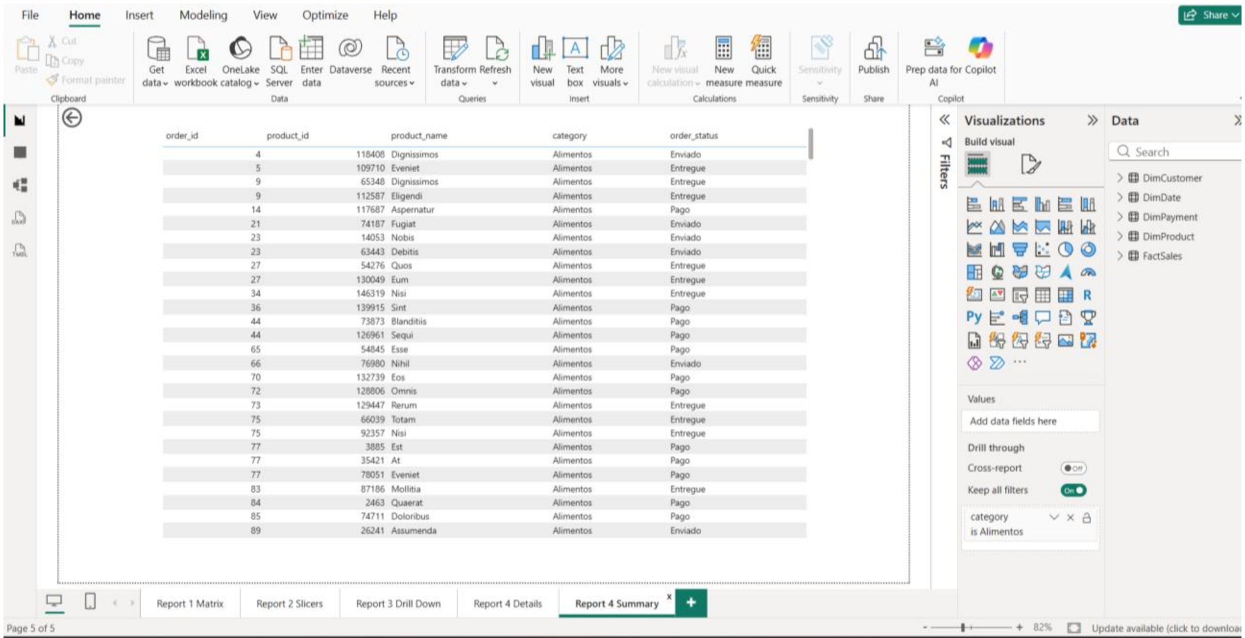
Report 3 demonstrates **hierarchical drill-down** functionality using a Clustered Column Chart. The X-axis is configured with a date hierarchy consisting of **Year → Quarter → Month**, and the Y-axis displays **Total Amount**. Users can drill down through the hierarchy by clicking on individual bars in the chart, allowing navigation from the yearly summary level down to quarterly and then monthly sales details. The drill-up button allows users to navigate back up the hierarchy. This report enables time-based trend analysis at multiple levels of granularity, supporting business decisions based on seasonal and monthly sales patterns.



Report 4

Report 4 implements a **drill-through** navigation feature across two pages. The first page (**Report 4 - Overview**) contains a Bar Chart displaying **Total Amount by Product Category**, providing a high-level summary of sales across all categories. The second page (**Report 4 - Detail**) contains a detailed Table visual showing **Order ID**, **Product Name**, **Product ID**, **Category**, and **Order Status** for individual transactions. The drill-through field is configured on the Category dimension, allowing users to right-click on any bar in the Overview chart and navigate directly to the Detail page, which automatically filters to show only the order-level data for the selected category. This feature enables users to seamlessly transition from summary-level insights to transaction-level details within the same report.





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